COURSE NAME: PLANT PROTECTION

COURSE CODE: NR 430

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- PATHOLOGY-

LECTURE 1:

1. INTRODUCTION

Plants cover the most area of the earth's living environment as trees, grasses, flowers, and so on. Plants play different important roles in the environment such as ecosystem balance and food supplement for animals and humans.

The study of plant pathogens belongs to the branch of biology known as **plant pathology**.

Diseases in plants are classified according to the two types of origins and these are **biotic and/or abiotic** origin. *Biotic (infectious)* diseases are developed owing to microbial infection, while abiotic (noninfectious) diseases are developed due to unfavorable environmental conditions, such as lack or excess of nutrients, moisture, light, and the presence of toxic chemicals in air or soil etc.

Plant pathology therefore focuses on the study of the organisms and the environmental factors that cause disease in plants; the mechanisms or methods by which these factors induce disease in plants and the methods of preventing or controlling disease aiming at reducing the damage it causes.

Essentially, plant pathology is correlated with other sciences such as entomology, bacteriology, mycology, virology, and weed science due to the visible effects of insects, bacteria, fungi, viruses/viroids, and weeds on plants, respectively

Some important terminologies:

- **Pathogen**: is defined as an organism causing disease to its host
- **Phytopathogens**: an organism which causes diseases in plants
- **Biotic diseases**: these are diseases caused by the pathogen (fungi, bacteria or viruses)

- **Abiotic diseases**: these are diseases which are caused by non-living agents like weather, soils, chemicals, or mechanical injuries. They are also referred to as non-infectious diseases.
- **parasitism:** an organism (parasite) lives on or in another organism (host) to obtain its required nutrition
- **Epiphytes**: any plant that grows upon another plant or object merely for physical support. Epiphytes have no attachment to the ground or other obvious nutrient source and are not parasitic on the supporting plants
- **Saprophytes:** An organism, especially a fungus or bacterium that lives on and gets its nourishment from dead organisms or decaying organic material. Saprophytes recycle organic material in the soil, breaking it down into in simpler compounds that can be taken up by other organisms.
- **pathogenicity**: this is the ability of an organism to cause a disease
- **Virulence:** is the disease producing power of an organism or ability of a pathogen or microbe to infect or damage a host. It can also be defined as the level of severity of the disease symptom.
- **Infectious diseases**: these are diseases that result from infection of a plant by a pathogen
- **Acquired resistance**: Plant resistance to disease activated after inoculation of the plant with certain microorganisms or treatment with certain chemical compounds.

2. THE CONCEPT OF DISEASE IN PLANTS

Unlike human beings and animals, it is not known whether plants also feel pain or discomfort and plants do not speak or otherwise communicate with us, it is difficult to pinpoint exactly when a plant is diseased especially if there are no visible signs.

A plant is considered to be healthy or normal, if it's able to carry out its physiological functions to the best of its genetic potential.

i. The meristematic (cambium) cells of a healthy plant should **divide and differentiate,**
ii. and different types of specialized cells (like the phloem) must be able to **absorb water and nutrients from the soil**; translocate these to all plant parts;
iii. a plant must be able to carry out **photosynthesis**, translocate, metabolize, or store the photosynthetic products;

iv. **and must be able to produce seed** or other reproductive organs for survival and multiplication.

Therefore, if the plant is unable to carry out most of these activities due to the presence of the pathogen or unfavorable environmental condition, then it is said to be **diseased**.

Plant disease, then, can be defined as the series of invisible and visible responses of plant cells and tissues to a pathogenic organism or environmental factor that result in adverse changes in the function, or integrity of the plant and may lead to partial impairment or death of plant parts or of the entire plant. The kinds of cells and tissues that become affected determine the type of physiological function that will be disrupted first. For example, infection of roots may cause roots to rot and make them unable to absorb water and nutrients from the soil.

The first step of plant disease treatment is observation of definite and clear symptoms on the plants. These symptoms give an initial indication for the type and cause of plant disease, which may end with the death

The infected part of the plant gives an indication of the type of plant disease, such as infected root which is usually correlated with root-rot disease.

Plant diseases can be classified and identified according to **several parameters** and some of them are:

i. disease symptoms,
ii. infected organ,
iii. infected plant type,
iv. and the type of phytopathogen.

Identifying the phytopathogen is considered to be more useful criterion used for plant disease classification, because it easily determines the disease cause, potential disease complications, and offer possible control methods.

Pathogenic microorganisms cause diseases in plants by;

i. Disturbing the metabolism of plant cells through enzymes, toxins, growth regulators, and other substances they secrete and by absorbing foodstuffs from the host cells for their own use.
ii. Some pathogens may also cause disease by growing and multiplying in the xylem or phloem vessels of plants, thereby blocking the upward transportation of water or the downward movement of sugars, respectively, through these tissues

Thousands of plant diseases exists with some affecting a lot of plant species while just a few are specific. It is important to note that a single plant species can also be affected by a hundred or more diseases.

LECTURE 2.

1. TYPES OF PLANT DISEASES AND BASIC PROCEDURES IN THE DIAGNOSIS OF PLANT DISEASES

The plant disease diagnosis depends on the exact determination of the disease cause. Generally, there are two plant disease causes: the **pathogens** and/or **environmental factors**. The former leads to **infectious** diseases, while the latter leads to **non-infectious** diseases.

The most common method of diagnosing a plant diseases is by grouping them according to the symptoms they cause

i. do they cause root rots, wilts, leaf spots, blights, rusts, smuts,
ii. the plant organ they affect (root diseases, stem diseases, foliage diseases),
iii. or to the types of plants affected (field crop diseases, vegetable diseases, tree diseases, turf diseases etc.

1.1. Infectious diseases

There are wide range of phytopathogens which cause infectious plant diseases such as fungi, bacteria, viruses, viroids, and protozoa. The infectious disease means the ability of phytopathogen to transfer diseases from the infected plant to another healthy one and causes the same disease and the same symptoms. Most phytopathogens can inhabit the internal environment of plants; however, some others can live on the plant surface such as some fungi, bacteria, and parasitic higher plants

Infectious, or biotic plant diseases

- Diseases caused by fungi
- Diseases caused by bacteria
- Diseases caused by parasitic higher plants and green algae
- Diseases caused by viruses and viroids
- Diseases caused by nematodes

Noninfectious, or abiotic plant diseases

- Diseases caused by too low or too high temperatures
- Diseases caused by lack or excess of soil moisture
- Diseases caused by lack or excess of light
- Diseases caused by lack of oxygen
- Diseases caused by air pollution
- Diseases caused by nutrient deficiencies

- Diseases caused by mineral toxicities
- Diseases caused by soil acidity or alkalinity (soil pH)

1.1.1. Diseases caused by parasitic higher plants

Some plant diseases are developed due to growing of certain plants attached on or in other plants, where they take all required nutrients without benefit sharing; these plants are called parasitic higher plants. This abnormal relationship leads to weakness of healthy or host plant. The parasitic higher plants are usually found attached with the surface of the host plant, such as dodder, witchweed, and broomrape.

1.1.2. Diseases caused by nematodes

The nematodes are one of most common phytopathogens which have definite symptoms. These symptoms only appear in the infected site. The nematode infections in or on plants are widely distributed especially in proper environments such as moderate temperature and high humidity.

2.1.3. Diseases caused by fungi

Interestingly, there are two main types of fungi appearing on plants: pathogenic and saprophytic. The pathogenic fungi live in or on plant tissues and cause serious complications for the vital physiological functions of plants, while saprophytic ones live in or on dead tissues. Accordingly, the diagnosis of plant disease must be exactly carried out. The exact diagnosis and determination of fungi take place by microscopical examination to identify the mycelial morphological characteristics, whatever fruiting structures and spores. After complete identification for the fungus and the symptoms of plant disease, the latter should be compared with that reported in the reference. This study will exactly determine whether the fungus is a pathogen or a saprophyte. Although microscopical examination is an essential and effective method for fungal identification, it only sometimes cannot lead to exact identification due to the absence of fungal fruiting structures and spores on infected plant tissue.

Therefore, an alternative method must be used, such as using selective media for isolation, identification, or promotion of sporulation. On the other hand, some fungi need to be incubated under certain temperature, aeration, or light conditions to produce spores

2.1.4. Diseases caused by bacteria and mollicutes

The appearance of bacterial growth in or on plant tissues means that bacterial plant disease may be present, because saprophytes may be present. Therefore, accurate bacterial identification must be carried out by using microscopical examination and physiological parameter determination.

The selective media are essentially used in the bacterial identification to determine the bacterial genus and species in some cases. Moreover, the confirmatory test of bacterial pathogenicity may be carried out by inoculation of single pure bacterial colony in the healthy plant, reproducing the same symptoms that appeared on the infected one. Moreover, immunodiagnostic techniques or serodiagnostic assays can be used, such as agglutination and precipitation, fluorescent antibody staining, and enzyme-linked immunosorbent assay (ELISA).

There are several advantages for these techniques such as quite sensitivity, fairly specificity, rapid, and easy to perform.

The molecular biological techniques are also widely used.

Mollicutes ;

There are uncommon microorganisms called **mollicutes and** similar to bacteria. These microorganisms are very small where they must be examined by an electron microscope. Mollicutes have polymorphism(2 or more variants of a particular DNA sequence) and lack cell wall–like mycoplasma. These microorganisms habit the young phloem cells as a convenient host, and cause severe plant diseases such as plant stunting, yellowing or reddening of leaves, proliferation of shoots and roots, production of abnormal flowers, and eventual decline and death of the plant. Mollicutes cannot be cultured on nutrient media except for the genus *Spiroplasma*. Mollicutes can be diagnosed by several parameters, such as symptoms determination, grafting, transformation, microscopical examination, susceptibility to tetracyclines, and so on.

2.1.5. Diseases caused by viruses and viroids

There are distinctive types of plant diseases caused by viruses/viroids. These diseases have definite and clear symptoms, which easily support disease diagnosis and are considered main advantage. Apart from this advantage, some recent techniques are widely used for disease diagnosis and virus identification, such as virus transmission tests to specific host plants by sap inoculation, grafting, certain insect, nematode, fungus, and mite vectors.

Moreover, serodiagnostic tests are used for this purpose such as enzyme-linked immunosorbent assays (ELISA), gel diffusion tests, micro-precipitin tests, and fluorescent antibody staining. The electron microscopy techniques as negative staining of virus particles in leaf dip or purified preparations are also used, as well as immune-specific electron microscopy.

On the other hand, there are more accurate techniques used for disease diagnosis and virus/viroid identification, such as electrophoretic tests and hybridization of commercially available radioactive DNA complementary to a certain virus DNA or RNA,

or viroid RNA, with the DNA or RNA present in plant sap and attached to a membrane filter (immunoblot).

2.1.6. Diseases caused by more than one pathogen

Sometimes, some plants are exposed to co-infection by two or more pathogens, which lead to the same or different disease symptoms. Therefore, the differentiation and identification of these pathogens are very essential to exactly determine the disease cause. The differentiation and subsequently identification are carried out by all techniques that are mentioned above.

2. PARASITISM AND PATHOGENICITY

The term parasitism called on the state in which an organism (parasite) lives on or in another one (host) to obtain its required nutrition. Usually, parasitism is correlated with **pathogenicity**, which means the ability of an organism to cause a disease.

Parasitism in some cases leads to a benefit relationship called **symbiosis**, in which both plant and organism alternate the benefits, such as bacterial nodules in the roots of legume plants and the mycorrhizal infection of feeder roots of most flowering plants. In the case of parasitism-pathogenicity relationship, the plant is diseased with the appearance of different symptoms such as increased respiration, disintegration or collapse of cells, wilting, abscission, abnormal cell division and enlargement, and degeneration of specific components such as chlorophyll.

The most common plant pathogens are fungi, bacteria, mollicutes, parasitic higher plants, parasitic green algae, nematodes, protozoa, viruses, and viroids. These parasites cause serious plant diseases, because they have the ability to penetrate the plant tissues to feed and proliferate in it, and withstand the conditions in which the host lives. These pathogens are also called **obligate parasites** because they can only live in their living hosts.

On the other hand, there are certain pathogens such as most fungi and bacteria can live on either living or dead hosts and on various nutrient media, so they are called **nonobligatory parasites**. Some nonobligatory parasites can grow saprophytically on dead organic matter, and therefore called semi-biotrophs/**facultative saprophytes**. There is a type of a life called facultative parasitism, in which an organism grows saprophytically (necrotrophs); however, under certain conditions, they attack living plants and cause a disease; these parasites are called **facultative parasites**. The type or degree of parasitism does not affect the disease severity. For instance, many diseases caused by weakly parasitic pathogens are much more damaging to a plant than others caused even by obligate parasites.

3. LOSSES CAUSED BY PLANT DISEASES

i. Plant Diseases Reduce the Quantity and Quality of Plant Produce

The type and amounts of losses caused by plant diseases vary with the plant, the pathogen, the locality, the environment, the control measures practiced, and combinations of these factors. The quantity of loss may range from slight to 100%. Plants or plant products may be reduced in quantity whilst in the field , or during storage.

ii. Plant Diseases May Limit the Kinds of Plants and Industries in an Area

Plant diseases may limit the kinds of plants that can grow in a large geographic area due to the effects of the disease dominant in that particular location.

iii. Plant Diseases May Make Plants Poisonous to Humans and Animals

Some diseases are very bad and make plant products unfit for human or animal consumption by contaminating them with poisonous fruiting structures.

iv. Plant Diseases May Cause Financial Losses

Financial losses from plant diseases can arise in many ways as farmers spend a lot in estabslishing farms. Therefore, plant diseases can cause losses and a farmer can lose the entire field in certain instances.

v. Economic loses

A farmer can experience serious loses as a result of reduce quality and quantity of the produce.

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LECTURE 3.

1. DEVELOPMENT OF DISEASE IN PLANT

The plant disease means the occurrence of physiological disorder(s) due to biotic agents such as microbial infection and/or abiotic agents such as extreme environmental factors.

In order for the plant disease to occur, an interaction must happen between two components: the plant and disease cause, which leads to physiological disorders. The disease cause is either biotic agent or abiotic agent as mentioned above. Interestingly, the biotic agents lead to infectious diseases, which develop under suitable environmental conditions. Therefore, the infectious diseases (occurred by pathogens) are not developed under extreme environmental conditions. This means it was impossible to get infectious and noninfectious plant diseases at the same time. The abiotic agents (environmental factors) play an important and vital role in the disease development and severity or disease resistance.

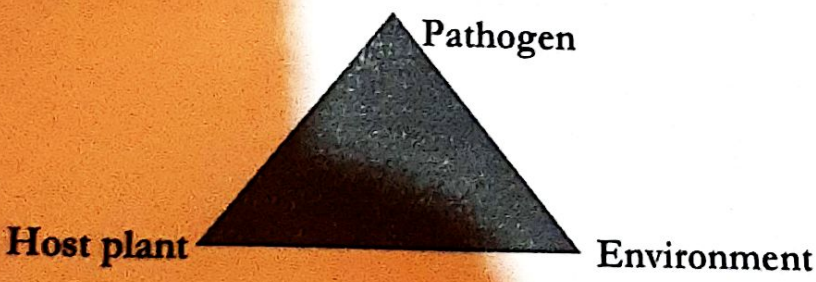
This matter depends mainly on different factors: the plant family, the plant age, plant genetic type, pathogen virulence race, pathogen inoculum size, and pathogen dormant state. Therefore, we can imagine the plant disease as a triangle, which is called "**disease triangle.**" The three sides of this triangle are the **plant, microorganisms**, and **the environmental factors**. The length of each side is proportional to the sum of the characteristics of the other two sides. For example, if the plant is resistant, the host side and the amount of disease would be small or zero, whereas if the plant is susceptible, the host side would be long and the potential amount of disease could be great .

1.1. Disease triangle

A plant becomes diseased in most cases when it is attacked by a pathogen or when it is affected by an abiotic agent. Therefore, in the first case, for a plant disease to occur, at least two components (plant and pathogen) must come in contact and must interact.

Additionally, a third component, namely a set of environmental conditions within a favorable range, must also occur for disease to develop.

The interactions of the three components of disease have often been visualized as a triangle and generally referred to as the "**disease triangle**." Each side of the triangle represents one of the three components.



I. Pathogen factors

- Virulence
- Inoculum type
- Dispersal potential
- Host range
- Alternation between hosts

II. Host plant factors

- Host resistance
- Age of the host plant
- Seasonal development
- Host uniformity

III. Environmental factors

- **Temperature:** Each pathogen has an optimum temperature for growth. In addition, different growth stages of fungi, such as the production of spores (reproductive units), their germination, and the growth of the mycelium (the filamentous main fungus body), may have slightly different optimum temperatures.

- **Light**
  Plants which receive adequate light will have less pathogen invesion as the conditions around them will be dry and physiological processes in the plants will be at their highest e.g photosynthesis.

- **Soil type**
  Certain pathogens are favoured by loam soils and others by clay soils. Usually, light soils that are low in organic matter are more preferred by pathogens.

- **Soil pH.**
  A measure of acidity or alkalinity, remarkebly influences a few diseases. Most pathogens thrive well in acidic conditions, hence need for soil neutralization through the additional of lime. Some pathogens may not survive high soil p H levels.

- **Moisture**
  High or low soil moisture may be a limiting factor in the development of certain root rot diseases. High soil-moisture levels favour development of destructive water mold fungi,. Excessive watering of palnts is a common problem. Overwatering, by decreasing oxygen and raising carbon dioxide levels in the soil, makes roots more susceptible to root-rotting organisms.
- **Humidity**
  High humidity favours development of the great majority of leaf and fruit diseases caused by fungi, water molds, and bacteria. Moisture is generally needed for spore germination, the multiplication and penetration of bacteria, and the initiation of infection.

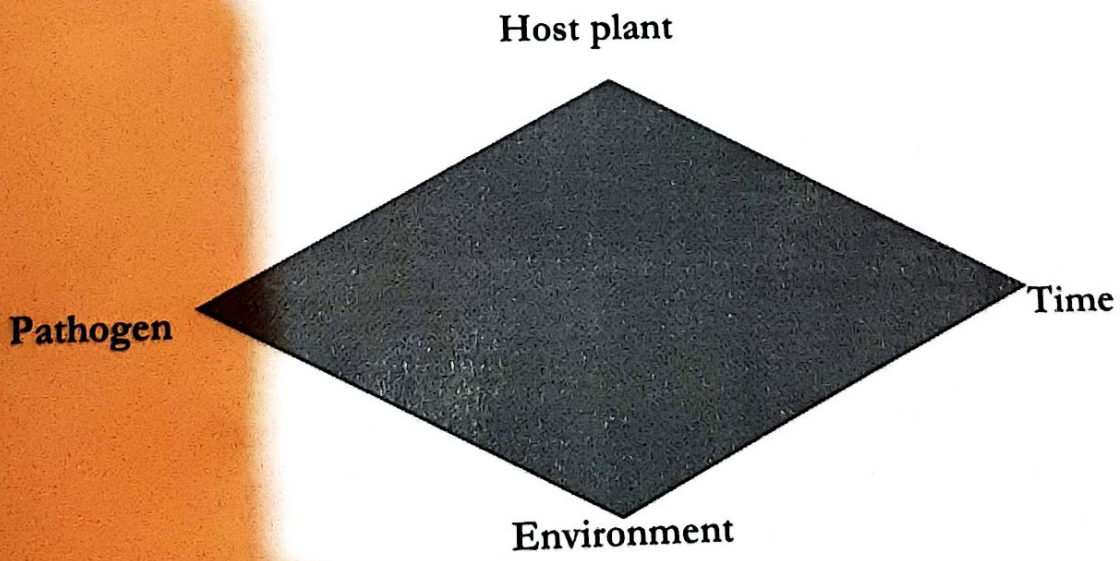
- **Air pollutant**

Pollution also influences disease development either by aiding pathogen development or in reducing their spread.

IV. Time

The **fourth factor** has been observed to have an influence in disease development is **'time'**.

Time factors include; season of the year, duration of favorable rains and temperature, appearance of the vector. The inclusion of time has made some scientist to stop refereeing to this interactions as disease triangle but ***''disease Tetrahedron''***.



1.2 PATHOGENESIS

Pathogenesis is the stage of disease in which the pathogen is in intimate association with living host tissue. **Three** fairly distinct stages are involved:

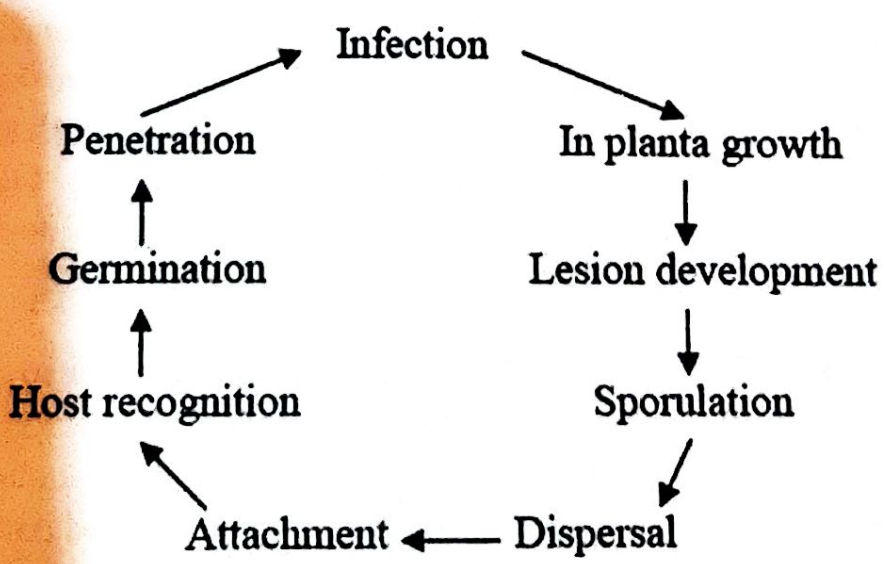
Inoculation: transfer of the pathogen to the infection court, or area in which invasion of the plant occurs (the infection court may be the unbroken plant surface, a variety of wounds, or natural openings—e.g., stomata [microscopic pores in leaf surfaces], hydathodes [stomata-like openings that secrete water], or lenticels [small openings in tree bark].

Incubation: the period of time between the arrival of the pathogen in the infection court and the appearance of symptoms

Infection: the appearance of disease symptoms accompanied by the establishment and spread of the pathogen.

1.3. DISEASE CYCLE

The disease cycle is a series of interconnected events, which lead to the disease development and pathogen propagation. These events include inoculation, prepenetration, penetration, infection, colonization (invasion), and growth and reproduction of the pathogen



The stages involved in disease cycle are outline below.

i. Inoculation

This is the transfer of the pathogen inoculum to the infection court, or area in which invasion of the plant occurs (the infection court may be the unbroken plant surface, a variety of wounds, or natural openings—e.g., stomata [microscopic pores in leaf surfaces], hydathodes [stomata-like openings that secrete water], or lenticels [small openings in tree bark]

In some cases, the inoculum is represented as an intact cell as in bacteria, mollicutes, protozoa, viruses, and viroids. There are two types of inoculum: **primary** and **secondary inoculum**, which in turn cause primary and secondary infection. The **primary inoculum** lives dormant in the winter or summer and causes the original infections in the spring or in the autumn.

The **secondary inoculum** is that one which is produced from primary infections. The primary inoculum is more abundant than secondary inoculum and closer to the crop, and caused more severe diseases and the losses that result.

ii. Pre-penetration

Attachment of pathogen to host;

Some pathogens directly penetrate the plant tissues by their vectors and then are surrounded by cytoplasm, cell membrane, or cell wall of plant cell, such as mollicutes, fastidious bacteria, protozoa, and most viruses. In other cases, the pathogen firstly makes contact with the external surface of the plant, and then penetration process takes place, such as fungi, bacteria, and parasitic higher plants. The adhesion of the pathogen with plant surface is carried out by mucilaginous substances found on the pathogen surface or at its tip. These substances are composed of mixture of water-insoluble polysaccharides, glycoproteins, and lipids, which, when moistened, become sticky and help the pathogen adhere to the plant.

iii. Penetration

Phytopathogens penetrate plant surfaces either through natural openings such as fungi and nematodes or through wounds in cell wall such as bacteria, viruses, viroids, mollicutes, fastidious bacteria, and protozoa. Penetration and infection are not usually correlated together, because some penetrated plants are resistant to phytopathogens [26].

iv. Spore germination

Spore germination process initiates by growth stimulation, which takes place with the availability of proper environmental conditions. Once the stimulation has been received,

the spore starts to utilize the stored food, such as lipids, and carbohydrates to build germ tube as abridge with cell membrane and cell wall of the plant. When appropriate physical and chemical signals, such as surface hardness, surface topography, and plant signals, are present, germ tube extension and differentiation take place

v. Recognition between host and pathogen

When a pathogen comes in contact with a host cell, the plant triggers a signal that either allows or retards the pathogen growth and development of disease. This signal is a biochemical reaction, which acts as a receptor to a pathogen contact. The pathogen propagation depends on the components of the plant cell, such as fatty acids galacturonan, phenolic compounds, amino acids, and sugars .

vi. Spores and seed germination

The availability and ability of host infection are increased by vegetative pathogen. The infection by fungal spores or parasitic higher plant seeds is carried out after germination has achieved. Fungal spores' germination is carried out by releasing either a mycelium or a germ tube that grows into the plant cell and cause infection.

vii. Growth of nematodes

The growth of nematodes starts with hatching of eggs, which essentially requires convenient environmental conditions such as temperature and moisture. After hatching of the eggs, the larvae penetrate the plant cell and grow to form the adults. After maturation, the adults of nematodes closely adhere with plant roots due to some factors, such as carbon dioxide and amino acids

viii. Infection

Infection is the process by which pathogens establish contact with susceptible cells or tissues of the host and procure nutrients from them. The intimate contact of phytopathogen with its host is called **infection process**. The infection process is either successful or unsuccessful depending on the type of host, whether susceptible or resistant, respectively. Successful infection results in the appearance of symptoms, such as discoloration, necrosis, dwarfism, and so on of the host.

While unsuccessful (latent) infection does not lead to any observations for the symptoms. As well known, the symptoms start to appear after the incubation period of the pathogen has been finished. The symptoms either are stable or may be changed until death .

1.3.1 Growth and reproduction of the pathogen

Most phytopathogens especially fungi and parasitic higher plants invade and infect plant tissues through the point of inoculation. Therefore, these pathogens can easily grow and

spread within the plant tissues until a certain limit or death occurs. For example, fungi can invade and infect the plant tissue to cause vascular wilts. This invasion is carried out by releasing spores within the vessels.

The time interval between inoculation and the appearance of disease symptoms is called the **incubation period**.

1.4 PARASITISM AND PATHOGENICITY

A parasite is an organism that lives on or in some other organism and obtains its food from there. The removal of food by a parasite from its host is called **parasitism**. Therefore, a plant parasite is an organism that becomes intimately associated with a plant and multiplies or grows at the expense of the plant.

Parasites removes water and nutrients from the host plant and reduces efficiency in the normal growth of the plant and becomes detrimental to the further development and reproduction of the plant.

In certain cases both the plant and the microorganism benefit from the association or relationship and this phenomenon is known as **symbiosis.**These parasites are successful because they can invade a host plant, feed and proliferate in it, and withstand the conditions in which the host lives.

Some parasites, including viruses, viroids, bacteria, nematodes, protozoa,and fungi that causes various diseases in plants like downy mildews, powdery mildews, and rusts, are called **biotrophs** (they can only live and multiply on another living organism) and are referred to as **obligate parasites**

Other parasites (most fungi and bacteria) can live on either living or dead hosts and on various nutrient media, and they are therefore called **non-obligate parasites**.

Some parasites spend most of the time and thrive well on dead organic matter **(necrotrophs)** but, under certain circumstances, may attack living plants and become parasitic; these parasites are called **facultative parasites.**

1.5 DISSEMINATION OF A PATHOGEN

A few pathogens, such as nematodes, fungi, and bacteria, can move short distances using their own power and thus can move from one host to another one very close to it.

However, there are several ways that contributes to the spread of pathogens from one area to another or form an infected plant to a health plant. These are:

i. By Vectors of Insects, Mites, Nematodes, Birds and Other Vectors

Insects, particularly aphids, leafhoppers, and whiteflies, are by far the most important vectors of viruses, whereas leafhoppers are the main vectors of bacteria, and protozoa. Each one of these pathogens is transmitted, internally, by only one or a few species of insects during feeding and movement of the insect vectors from plant to plant.

The insects carry these pathogens externally from plant to plant and deposit them on the plant surface or in the wounds they make on the plants during feeding.

ii. Dissemination by Humans

Human beings disseminate all kinds of pathogens over short and long distances in a variety of ways. Within a field, humans disseminate some pathogens, such as tobacco mosaic virus, through the successive handling of diseased and healthy plants. In addition, human beings may also carry pathogens on the clothing and working shoes and introduce in another area.

iii. Dissemination by air

Spores of most oomycetes (a subclass of parasitic fungi that includes water molds, white rusts, and downy mildews) and most fungi and the seeds of most parasitic plants are disseminated by air currents that carry them as inert particles over long distances.

iv. Dissemination by water

Water is important in disseminating pathogens in three ways. Firstly, Bacteria, nematodes, and spores of fungi present in the soil are disseminated by rain or irrigation water that moves on the surface or through the soil. Secondly, all bacteria and the spores are spread by the rain or (overhead) irrigation water, which either washes them downward or splashes them in all directions.

Thirdly, raindrops or drops from overhead irrigation pick up the fungal spores and any bacteria present in the air and wash them downward, where some of them may land on susceptible plants.

v. Dissemination by farming implements

Pathogens can also be disseminated through tools, such as contaminated pruning equipment used on diseased plants and then carried to healthy plants. Farming equipments such as tractors and hoes can aid the transportation of pathogens from infected arrears to healthy areas.

vi. Using infected planting materials

Infected transplants, cuttings, suckers, seed, and nursery stock when transported from one region to another can disseminate pathogens.



LECTURE 4.

MARCH, 2022

1.0 HOW PLANTS DEFEND THEMSELVES AGAINST PATHOGENS

It is very important to understand that whatever kind of defense or resistance a host plant employs against a pathogen or against an abiotic agent, it is ultimately controlled, directly or indirectly, by the genetic material (genes) of the host plant and that of the pathogen. Therefore, the following are some of the mechanisms plants uses to defend themselves.

i. Non-host Resistance

A plant may find it easy to defend itself against pathogens when it is brought in contact with a pathogenic biotic agent to which the plant is not a host. This is known as non-host resistance and is the most common form of resistance. For example, apple trees are not affected by pathogens of potatoes, tomato, maize and so on because the genetic makeup of apple is in some way(s) different from that of host plants, which, of course, they are attacked by their own pathogens.

ii. Preexisting Defense Structures

The first line of defense of a plant against pathogens is its surface, which the pathogen must adhere to and penetrate if it is to cause infection. Some structural defenses are present in the plant even before the pathogen comes in contact with the plant. Some of them are

-structure and thickness of the epidermal cell walls (thick walls makes direct penetration by fungal pathogens difficult or impossible

-size, location, and shapes of stomata (if more stomata are located on the underside of the leaf then few vectors will have access, also varieties that have stomata that do not open much or open for a shorter period offer some form of defense as the pathogen cannot penetrate)

--quality of wax (forms a water repellent surface that prevent the pathogen to be deposited)

- thick cuticle (increase resistance to infection in diseases in which the pathogen enters its host only through direct penetration)

-presence of tissues made of thick-walled cells that hinder the advance of the pathogen on the plant

- **Hairy leaf and stem surface** (prevents the vectors to land there and reduce and prevent the pathogens to be deposited)

-**presence of thorns** (this also deters the vector to land on such plants)

iii. Preexisting Chemical Defenses

A particular pathogen will not infect certain plant varieties even though no structural barriers of any kind seem to be present or to form in these varieties. This clearly suggest that defense mechanisms of a chemical rather than a structural nature are responsible for the resistance to infection exhibited by plants against certain pathogens.

- **Inhibitors Released by the Plant in Its Environment:** Certain plants release a variety of substances through the surface of their aboveground parts as well as through the surface of their roots. Some of the compounds released by certain kinds of plants, however, seem to have an inhibitory action against certain pathogens. Fungitoxic exudates on the leaves of some plants, e.g., tomato and sugar beet, seem to be present in sufficient concentrations to inhibit the germination of spores of fungi

- **Inhibitors Present in Plant Cells before Infection:** some plants are resistant to diseases caused by certain pathogens because of one or more inhibitory antimicrobial compounds, known as ***phytoanticipins***, which are present in the cell before infection.

iv. Defense through lack of essential factors

-Lack of Recognition between Host and Pathogen: Plants of a species or variety may not become infected by a pathogen if their surface cells lack specific recognition factors (specific molecules or structures) that can be recognized by the pathogen. If the pathogen does not recognize the plant as one of its host plants, it may not become attached to the plant or may not produce infection substances.

-Lack of Host Receptors and Sensitive Sites for Toxins: the toxin, which is responsible for the symptoms, is thought to attach to and react with specific receptors or sensitive sites in the cell. Only plants that have such sensitive receptors or sites become diseased. Plants of other varieties or species that lack such receptors or sites remain resistant to the toxin and will develop no symptoms.

- **Lack of Essential Substances for the Pathogen:** plants that for some reason do not produce one of the substances essential for the survival of an obligate parasite would be resistant to the pathogen that requires that subatance for survival.

v. Induced structural defenses

Plants usually respond by forming one or more types of structures that are more or less successful in defending the plant from further pathogen invasion even after penetration of the pathogen through wounds and natural openings like stomata.

-Some of the defense structures formed involve the cytoplasm of the cells under attack where the cytoplasm becomes dense and disintegrates the pathogen, and the process is called ***cytoplasmic defense reaction***;

-some defense involve the walls of invaded cells which undergoes morphological changes and are called ***cell wall defense structures***.

-Further, death of the invaded cell may protect the plant from further invasion. This is called the ***necrotic*** or ***hypersensitive defense reaction.***

-Formation of Abscission Layers: this is the layer formed on young, active leaves of stone fruit trees after infection by any of several fungi, bacteria, or viruses. The plant will discard the infected area along with a few yet uninfected cells, protects the rest of the leaf tissue from being invaded.

-Deposition of the gums: The defensive role of gums stems is that they are deposited quickly in the intercellular spaces and within the cells surrounding the region of infection, thus forming an impenetrable barrier that completely encloses the pathogen. The pathogen then becomes isolated, starves, and sooner or later dies.

vi. Genetically engineering disease-resistant plants

The number of plant genes for resistance (R genes) that have been isolated is increasing rapidly across the world. Scientist have managed to isolate and make certain plant species more resistant to certain pathogens.

2. CULTURAL PRACTICES AND CONTROL MEASURES AIM AT REDUCING PLANT DISEASES

i. Site Selection and Preparation

Low-lying and poorly drained and aerated fields, especially if near other infected fields, tend to favor the appearance and development of epidemics.

ii. Selection of Propagative Material

The use of seed, nursery stock, and other propagative material that carries various pathogens increases the amount of initial inoculum within the crop and favors the development of epidemics greatly.

The use of pathogen-free or treated propagative material therefore, can reduce the chance of epidemics greatly.

iii. Cultural Practices

Continuous monoculture, large hectors planted with the same variety of crop, high levels of nitrogen fertilization, no-till culture, dense plantings, overhead irrigation, injury by herbicide application, and poor sanitation all increase the possibility and severity of epidemics because the conditions become very favorable for disease development.

iv. Disease Control Measures

Chemical sprays, cultural practices (such as sanitation and crop rotation), biological controls (such as using resistant varieties), and other control measures reduce or eliminate the possibility of a disease.

v. Introduction of new Pathogens

The frequency of worldwide travel have also increased the movement of seeds, tubers, nursery stock, and other agricultural goods. These activities increase the possibility of introducing pathogens into areas where the hosts have not had a chance to evolve resistance to these pathogens. Such pathogens frequently lead to severe epidemics.

vi. Inspection and quarantine

Rigorous inspection measures have been put in place by authorities to ensure that only seeds and plant products that meet standards are allowed to reach the market or entry into the country or region.

vii. Crop Certification

Several voluntary and compulsory inspection systems are in effect in various countries to inspect and issue certificates for seed and plant products. Only those which are certified and tested reach the market and stiff penalties are in place to those who evade the inspection and certification process.